

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I Prudhvi Raj.B(192324274) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prudhvi raj.b

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks) Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

6. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models. (5 Marks)

7. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making. (5 Marks)

8. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance. (5 Marks)

9. Consider the application of **Genetic Algorithms for feature selection** in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity. (5 Marks)

10. In the context of **classification problems**, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection. (5 Marks)

11. With the help of a neat diagram, explain the architecture of an **Artificial Neural Network (ANN)**. Describe the roles of input, hidden, and output layers, and explain how weights and biases are used in computation. Illustrate how forward propagation works with a simple example, and discuss how overfitting can be controlled using techniques such as dropout and regularization. (5 Marks)

12. Design a machine learning system for **weather forecasting** to predict rainfall using features such as temperature, humidity, wind speed, and atmospheric pressure. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time predictions. (5 Marks)

13. In the context of **Recurrent Neural Networks (RNNs)**, explain the architecture and how they handle sequential data. Describe the concept of hidden states and how information is passed through time steps. Illustrate with an example such as text prediction or time-series forecasting, and discuss challenges like vanishing gradients and possible solutions. (5 Marks)

14. Consider the use of **Genetic Algorithms for optimization problems** such as scheduling tasks in a system. Explain how solutions are encoded as chromosomes and how the fitness function is defined. Describe the processes of selection, crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently. (5 Marks)

15. In the context of **classification models**, explain how performance evaluation is carried out using metrics such as accuracy, precision, recall, F1-score, and ROCAUC. Using a sample confusion matrix, demonstrate how each metric is calculated and discuss their importance in real-world applications such as spam detection and disease diagnosis. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Scenario-Based Questions – Answers

Question 1

Functional blocks of a machine learning system, illustrated with a neat sketch, and examples for supervised and unsupervised learning.

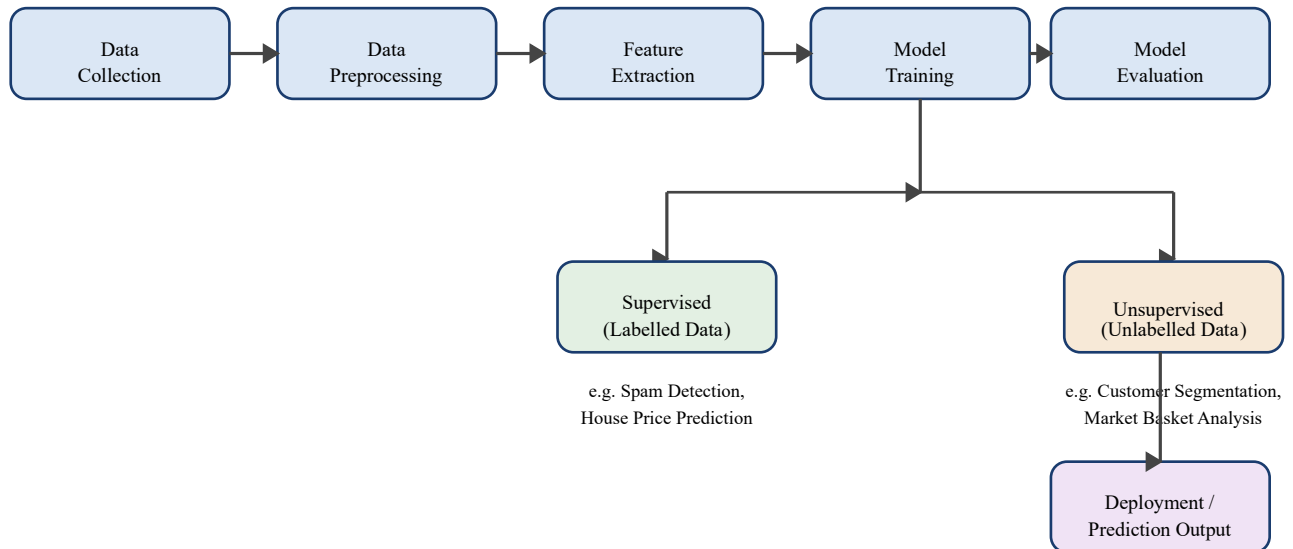


Fig. 1: Functional block diagram of a Machine Learning System

Functional blocks of the ML system:

- **Data Collection:** Gathering raw data from relevant sources such as sensors, databases, surveys, or the web. The quality and quantity of collected data directly affects the model's performance.
- **Data Preprocessing:** Cleaning the raw data by handling missing values, removing duplicates/outliers, normalizing or scaling numeric values, and encoding categorical attributes so it is suitable for modelling.
- **Feature Extraction:** Identifying and constructing the most relevant attributes (features) from the preprocessed data, using techniques such as PCA, correlation analysis, or domain-based feature engineering, to improve model efficiency.
- **Model Training:** Feeding the prepared features into a learning algorithm so it can learn the underlying patterns/relationships in the data. This is where the distinction between supervised and unsupervised learning arises.
- **Model Evaluation:** Assessing the trained model's performance on unseen (validation/test) data using metrics such as accuracy, precision, recall, or clustering quality, before deployment.

Supervised learning (labelled data): Each training example is paired with a known output/label. The model learns a mapping from inputs to outputs by comparing its predictions with the actual labels and

minimizing the error. *Example:* Predicting house prices from features such as area, location, and number of rooms, where the actual sale price is the label used to train and correct the model.

Unsupervised learning (unlabelled data): The training data has no output labels; the model must discover hidden structures or groupings on its own. *Example:* Customer segmentation, where a clustering algorithm (e.g., K-Means) groups customers with similar purchasing behaviour without being told in advance what the groups should be.

In both cases the data flows through the same collection, preprocessing, and feature-extraction stages; the difference lies in whether labels are available at the training stage, which determines whether the algorithm learns by matching predictions to known answers (supervised) or by finding patterns/similarities in the data itself (unsupervised).

Question 2

Design a machine learning system for disaster management (floods, earthquakes) using historical and real-time data.

(a) System overview: The system continuously ingests historical records (past flood/earthquake events, rainfall, river levels, seismic activity) together with real-time data streams (weather radar, IoT river/soil-moisture sensors, seismographs, satellite imagery) to predict the likelihood, location, and severity of an impending disaster.

(b) Model selection and justification:

- **Flood prediction:** A **Recurrent Neural Network / LSTM** is well suited because rainfall, river-level, and soil-moisture readings form sequential time-series data, and LSTMs can capture long-term temporal dependencies (e.g., cumulative rainfall effect over several days) better than traditional models.
- **Earthquake prediction / risk mapping:** A combination of a **Convolutional Neural Network (CNN)** for satellite/seismic-image data and an **ensemble model (e.g., Random Forest / Gradient Boosting)** for tabular sensor features is appropriate, since ensembles handle noisy, heterogeneous sensor data robustly and give interpretable feature importance, which is valuable for risk assessment.

(c) Data preprocessing steps:

- Handling missing/faulty sensor readings via interpolation or imputation.
- Removing noise and outliers caused by sensor malfunction.
- Normalizing/scaling variables (rainfall in mm, seismic magnitude, water level in m) onto a common scale.
- Synchronizing and time-aligning multi-source data (historical + real-time streams) into a unified time series.

(d) Feature selection techniques: Correlation analysis and mutual information to remove redundant sensor variables; Principal Component Analysis (PCA) to reduce dimensionality of large satellite/sensor feature sets; and domain-driven engineered features such as rainfall intensity rate, rate-of-change of river level, and seismic wave amplitude/frequency ratios.

(e) Training, validation and evaluation: The historical labelled dataset (past events marked as disaster / no-disaster) is split into training, validation, and test sets (e.g., 70:15:15), often using time-

based splitting to avoid data leakage. The model is trained by minimizing a suitable loss function (e.g., binary cross-entropy), with hyperparameters tuned on the validation set. Because disaster events are rare (imbalanced classes), evaluation relies on precision, recall, F1-score, and ROC-AUC rather than accuracy alone, since recall (catching true disaster events) is especially critical.

(f) **Assisting authorities:** The trained system can generate real-time **early-warning alerts** when predicted risk crosses a threshold, produce **risk maps** highlighting vulnerable regions for proactive planning, and support **emergency response** by prioritizing resource allocation (evacuation routes, relief materials, medical teams) to the areas of highest predicted impact — ultimately reducing response time and potential loss of life.

Question 3

Backpropagation in a multilayer perceptron for binary classification: forward pass, error calculation, backward propagation, and the influence of learning rate and activation function.

Example network: One input ($x = 1$), one hidden neuron (weight $w_1 = 0.4$, bias $b_1 = 0.1$), one output neuron (weight $w_2 = 0.6$, bias $b_2 = -0.2$), sigmoid activation, target output $t = 1$, learning rate $\eta = 0.5$.

Step 1 – Forward pass:

Hidden layer input: $h_{in} = (w_1 \times x) + b_1 = (0.4 \times 1) + 0.1 = 0.5$

Hidden layer output: $h_{out} = \text{sigmoid}(0.5) = 1 / (1 + e^{-0.5}) \approx 0.6225$

Output layer input: $o_{in} = (w_2 \times h_{out}) + b_2 = (0.6 \times 0.6225) - 0.2 \approx 0.1735$

Output: $o_{out} = \text{sigmoid}(0.1735) \approx \mathbf{0.5433}$

Step 2 – Error calculation:

Error = Target – Output = $1 - 0.5433 = \mathbf{0.4567}$

Step 3 – Backward propagation of error:

Output layer gradient: $\delta_{out} = \text{Error} \times o_{out} \times (1 - o_{out}) = 0.4567 \times 0.5433 \times 0.4567 \approx 0.1133$

Update w_2 : $w_2(\text{new}) = w_2 + \eta \times \delta_{out} \times h_{out} = 0.6 + (0.5 \times 0.1133 \times 0.6225) \approx \mathbf{0.6353}$

Hidden layer gradient: $\delta_{hidden} = \delta_{out} \times w_2 \times h_{out} \times (1 - h_{out}) \approx 0.1133 \times 0.6 \times 0.6225 \times 0.3775 \approx 0.0160$

Update w_1 : $w_1(\text{new}) = w_1 + \eta \times \delta_{hidden} \times x = 0.4 + (0.5 \times 0.0160 \times 1) \approx \mathbf{0.4080}$

Conclusion: After one backpropagation iteration, the weights are updated from ($w_1=0.4$, $w_2=0.6$) to approximately ($w_1=0.408$, $w_2=0.635$), moving the network's output closer to the target.

Influence of learning rate: A small learning rate leads to slow but stable convergence, requiring many epochs; a very large learning rate can cause the weight updates to overshoot the minimum, leading to oscillation or divergence instead of convergence. A well-tuned learning rate balances convergence speed with stability.

Influence of activation function: Sigmoid/tanh give smooth, differentiable outputs but suffer from the vanishing-gradient problem in deep networks, slowing learning in early layers. ReLU and its variants avoid vanishing gradients for positive inputs and generally allow faster convergence, but poor initialization can cause “dead neurons.” The choice of activation function therefore directly affects how quickly and accurately the network converges during training.

Question 4

Genetic Algorithm evolution over Speed/Accuracy dataset – selection, crossover, mutation, and the final optimal solution.

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Step 1 – Selection: The Genetic Algorithm selects individuals with the highest fitness values to act as parents for the next generation (analogous to roulette-wheel or tournament selection, where higher fitness gives a higher chance of being chosen). Here, **Example 5 (Medium Speed, High Accuracy, fitness = 80)** and **Example 3 (High Speed, Medium Accuracy, fitness = 75)** are the fittest individuals and are selected as parents, since both are marked “Suitable = Yes.”

Step 2 – Crossover: The Speed and Accuracy “genes” of the two selected parents are exchanged to produce new offspring that combine favourable traits from both parents:

Parent	Speed	Accuracy
Parent A (Ex.5)	Medium	High
Parent B (Ex.3)	High	Medium
Offspring 1 (crossover)	High	High
Offspring 2 (crossover)	Medium	Medium

Offspring 1 combines the best trait of each parent (High Speed from Parent B, High Accuracy from Parent A), giving it the highest expected fitness (≈ 90) among the new generation.

Step 3 – Mutation: To maintain diversity and explore the hypothesis space beyond what crossover alone can reach, a random gene is mutated with a small probability — for example, mutating Offspring 2's Speed from Medium to High produces a new individual (High Speed, Medium Accuracy), preventing the population from converging prematurely and allowing exploration of combinations not present in the original parents.

Step 4 – Evolution across generations:

Generation	Best Individual	Speed	Accuracy	Fitness
Generation 0 (initial)	Example 5	Medium	High	80
Generation 1 (after crossover)	Offspring 1	High	High	≈ 90
Generation 1 (after mutation)	Mutated Offspring 2	High	Medium	≈ 78

Final optimal solution: After selection, crossover, and mutation, the population converges towards High Speed and High Accuracy (Offspring 1, fitness ≈ 90) as the best optimal, “Suitable = Yes” solution, since it maximizes the fitness function more than any individual in the original population.

Question 5

Neural network for disease classification: training via backpropagation and evaluation using the given dataset of Actual vs Predicted outputs.

Ex.	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual	Predicted
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Training with backpropagation: The categorical attributes (Blood Pressure, Glucose Level, Lifestyle Risk) are encoded numerically (e.g., Low=0, Medium=1, High=2) and combined with the numeric Age feature to form the input vector for each example. In the forward pass, the network computes a predicted probability of disease presence through its hidden and output layers. This prediction is compared with the actual output (Yes/No) using a loss function such as binary cross-entropy, and the resulting error is propagated backward to update the weights (as illustrated in Question 3), gradually reducing the error over multiple training epochs.

Step 1 – Identify TP, TN, FP, FN by comparing Actual vs Predicted:

Ex.	Actual	Predicted	Category
1	Yes	Yes	True Positive (TP)
2	Yes	No	False Negative (FN)
3	No	No	True Negative (TN)
4	Yes	Yes	True Positive (TP)
5	No	Yes	False Positive (FP)

Counting: **TP = 2, TN = 1, FP = 1, FN = 1** (Total = 5 examples)

Step 2 – Evaluation metrics:

Metric	Formula	Calculation	Result
Accuracy	$(TP+TN)/Total$	$(2+1)/5$	60%
Precision	$TP/(TP+FP)$	$2/3$	66.67%
Recall (Sensitivity)	$TP/(TP+FN)$	$2/3$	66.67%
Specificity	$TN/(TN+FP)$	$1/2$	50%
F1-Score	$2 \times (P \times R) / (P + R)$	$2 \times (0.6667 \times 0.6667) / 1.3334$	66.67%

Performance analysis: The model correctly classifies 3 out of 5 patients (60% accuracy), with equal precision and recall (66.67%), indicating a moderate but not highly reliable performance on this small dataset. The false negative (Example 2, an actual disease case missed) is particularly concerning in a medical context, as it could delay treatment, while the false positive (Example 5) may cause unnecessary patient anxiety or further testing.

How preprocessing and feature selection improve accuracy and reliability: Preprocessing steps such as normalizing Age, consistently encoding categorical attributes (Blood Pressure, Glucose Level,

Lifestyle Risk), and handling any missing values ensure the network receives clean, comparable inputs rather than being biased by scale differences. Feature selection techniques (e.g., correlation analysis or feature importance ranking) help identify which attributes (such as Glucose Level or Blood Pressure) are most predictive of the disease, removing noisy or redundant features. Together these steps reduce overfitting, help the network converge faster during backpropagation, and generally improve the accuracy, precision, and recall of the model on unseen patient data.